A Voluntary Price Cap in the Italian Retail Fuel Market:
Adoption, Counterfactual Prices, and Inflation

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Abstract

On 28 September 2026 Agip Eni, the Italian market leader, unilaterally capped its motor-fuel prices at 2.00 €/l (petrol) and 2.20 €/l (diesel) for two months, inviting competitors to follow. We reconstruct the daily effective price of every Italian motorway-free (*stradale*) fuel station from the official price-communication registry and ask four questions: who adopted the cap, how adoption diffused, what prices would have been without it, and what it did to measured inflation. Within fifteen days, 70.2% of 39,423 station–fuel units priced at or below the (fuel- and date-adjusted) cap on at least one day (any-day shares: 62.5% petrol, 78.0% diesel; median adoption: day 3). Cox duration models show adoption is faster for stations with a smaller pre-cap gap to the threshold (hazard ratio 1.4×10^{-2} per €/l), higher local population density, larger chains, and smaller distances to the nearest competitor (-9% per km) and to the nearest adopter (-2 to -5% per km, stronger once the leader brands are excluded); a station–fixed-effects event study confirms a spatial diffusion gradient with no anticipation. The cap polarised rather than compressed the market: province-level within-day dispersion roughly doubled in 214 of 220 province–fuel cells. To measure the consumer-price effect we build a counterfactual: an autoregressive distributed-lag model links the national net-of-tax price of each fuel to the Brent crude price in euro over the pre-cap window (1 July–27 September); projected forward on the actual oil path, it predicts an October pump price 16.5 c/l (petrol) and 27.5 c/l (diesel) above the realised one on average. On the ISTAT reference days used for inflation measurement, month-on-month fuel-price changes were -2.3% (petrol) and $+1.6\%$ (diesel) with the cap, against $+4.9\%$ and $+13.4\%$ without it: the cap lowered October's headline month-on-month inflation by roughly 0.43 percentage points. Because daily price extracts close at 8am, a communication wave crossing a sampling reference day additionally biases recorded averages by up to 4 c/l; communication-time reconstruction corrects it.

1 Introduction

Price caps are among the bluntest instruments of price regulation, and their effects are notoriously hard to measure: where a cap binds, prices cluster at the ceiling and observed dispersion collapses, confounding changes in conduct with censoring. On 28 September 2026, Italy offered an unusual natural experiment. Agip Eni, the market leader, announced a *voluntary* two-month cap on its retail motor-fuel prices (2.00 €/l for petrol, 2.20 €/l for diesel, self-service), framed as a measure against speculation. No law mandated it; competitors were simply invited to match. Within days, the largest synchronised price- communication wave in the market's recorded history followed: on 1 October alone, 9,192 stations communicated new petrol prices, led by rival chains cutting 10–17 c/l overnight.

The announcement landed on an unusual macroeconomic backdrop. Brent crude — our reference for the marginal cost of motor fuels — rose by three quarters in euro terms over our window (an index of 100 on 1 July reaches 173 on the cap date), so October pump prices were being pushed upward by world markets at the very moment the cap was pushing them down. Any statement that "the cap lowered prices" is therefore incomplete without a counterfactual: the observation that prices rose less than oil says something, but how much less, and with what consequence for measured inflation? This paper reconstructs the whole chain: from the diffusion

of adoption across stations, to the price path the market would have followed absent the cap, to the contribution of that difference to official consumer-price inflation.

We ask four questions. First, *who* adopted the cap: which station and firm characteristics predict pricing at or below the threshold? Second, *how did adoption diffuse* across the network of stations: does adoption of neighbouring stations raise a station's own adoption hazard, as strategic-complementarity models would predict? Third, *what would prices have been without the cap*: conditional on the oil price and the pre-cap pass-through dynamics, what net-of-tax prices — and hence pump prices — would October have delivered? Fourth, *what did the cap do to measured inflation*: on the reference days used by the national statistical institute, what month-on-month change occurred, and what would have occurred in the counterfactual?

We answer these questions with a station-level panel reconstructed from the MIMIT price-communication registry. The registry records every price communication (`dtComu`) with date and time, which lets us build the *effective* daily price path of each station — the last communicated price as of each day — rather than the daily snapshot published each morning. This distinction is not academic: the daily extracts stop at 8am, so a station communicating a lower price at 7am is only recorded the following day. On ISTAT price-reference days (the 1st, 11th, and 21st of each month) this biases the recorded national average: for 1 October 2026 we measure an overstatement of 3.5–4.1 c/l (section 4).

Our setting has two attractive features. Because the cap was *voluntary*, adoption is a firm decision, and we can measure the diffusion of a coordinated price cut — not just the mechanical effect of a legal ceiling. And because we observe the communication time, adoption dates are exact to the day rather than contaminated by reporting lag. For the counterfactual, the relevant margin is the *net-of-tax* price — what the market keeps after excise and VAT are stripped out — because the tax calendar itself moved violently during 2026: the diesel excise alone swung by up to 14 c/l between July and October (section 2). Modelling net prices makes the counterfactual immune to these tax interventions and lets us re-apply the actual tax calendar to both the observed and the counterfactual path, so that tax steps contaminate neither side of the comparison.

We find that adoption was fast but far from universal: 70.2% of station–fuel units priced at or below the cap at some point in the first fifteen days (final-day shares: 52.9% petrol, 70.3% diesel; median: day 3), and adoption clustered along urban, organised-firm corridors. Hazard models (section 6) show the pre-cap gap to the threshold dominates all other covariates; distance to the nearest adopter, population density and chain size all raise the adoption hazard significantly. The event study (section 7) shows no anticipation and a clean diffusion gradient across day-0 adopter-distance terciles. The cap also *polarised* the market (section 8): within-province dispersion widened in almost every province–fuel cell, and non-adopters became systematically more expensive relative to their provincial market.

On the counterfactual (section 9), an ARDL model of the daily national net price on lags of itself and of the Brent price in euro — estimated only on pre-cap data, and validated by a placebo backtest that halves the random-walk forecast error for petrol — implies October pump prices 16.5 c/l (petrol) and 27.5 c/l (diesel) above the realised ones on average over 28 September–31 October, and 20.7 and 31.8 c/l above by the window's end. Translated to measurement days (section 10): the realised month-on-month change of pump prices averaged over the 1st, 11th and 21st was −2.3% for petrol and +1.6% for diesel; the counterfactual delivers +4.9% and +13.4%. Using the 2026 ISTAT basket weight for motor fuels (4.52%), the cap lowered headline month-on-month inflation by roughly 0.43 percentage points in October — the difference between a +0.41 pp fuel contribution and a −0.02 pp one.

The paper proceeds as follows. Section 2 describes the institutional setting: the cap, the tax calendar, and how official inflation statistics sample fuel prices. Section 3 positions the paper in the literature. Section 4 describes the data and the communication-based price reconstruction, including the oil and tax series. Section 5 presents descriptive evidence. Sections 6 and 7

analyse adoption (survival and panel designs). Section 8 studies dispersion and competitiveness. Section 9 builds the oil-based price counterfactual, and Section 10 maps it to measured inflation. Section 11 discusses policy implications and Section 12 concludes.

2 Institutional background

The market and the registry. Italian fuel retailers must communicate every price change to the MIMIT (Ministry of Enterprise and Made in Italy) price-communication registry, which publishes daily extracts by station, fuel and price with the communication timestamp. Pump prices are liberalised; the communicated price is the legally displayed price. Motorway (*autostradale*) and ordinary-road (*stradale*) stations are distinct markets with different cost levels; this paper restricts to *stradale* and to *self-service*, the mode used by the vast majority of purchases.

The voluntary cap. On 28 September 2026 Agip Eni — the brand with the largest network — announced it would cap its self-service prices at 2.00 €/l (petrol) and 2.20 €/l (diesel) for two months, and invited competitors to match. The cap is a private commitment: no law mandates it, no authority enforces it, and each rival's compliance is its own decision. Because the announced thresholds sit close to the pre-cap national average petrol price but clearly above the diesel excise-driven level, the two fuels provide different margins of bite. The announced diesel cap was framed on pre-tax-change levels: when the government removed the remaining 5 c/l excise discount on diesel with effect from 6 October 2026, the corresponding "compliant" diesel threshold moved to $2.20 + 5 \times 1.22 = 2.261$ €/l, the mechanical translation of a tax change through 22% VAT.

The 2026 excise calendar. Excise duties (accise) are set per litre and enter the pump price additively; VAT of 22% applies multiplicatively to the sum of net price and excise, so $P^{gross} = (P^{net} + \tau) \times 1.22$. In 2026 the government adjusted excises repeatedly. Within our window: petrol excise rose from 62.29 to 67.29 c/l on 4 July; diesel followed the same step on 4 July, was cut by 14 c/l on 28 July (to 53.29 c/l), then raised to 57.29 (18 September), 62.29 (26 September), 67.29 (6 October, remaining discount removed by decree-law 162/2026) and cut again to 63.29 c/l on 28 October. These are movements of the same order as — sometimes larger than — the daily dynamics the cap was meant to control, and a pump-price series is uninterpretable without them. All counterfactual comparisons in this paper therefore work with net prices, and re-apply the actual excise calendar identically to both sides.

How official statistics measure fuel prices. ISTAT collects fuel prices for the consumer price index (NIC) on the 1st, 11th and 21st of each month, the three *reference days* whose monthly average enters the index. Motor fuels (ECOICOP class 07.2.2, "Carburanti e lubrificanti") carry 4.5219% of the 2026 NIC basket (definitive weights). Two features matter for us. First, the measured month-on-month change is the difference between the average of the three October reference days and that of the three September days, so a policy that bites mid-month is captured with a lag and a blend — here, the cap starts nine days before the first October reference day. Second, the collection is timed on the day's registry state, which — because extracts close at 8am — can mis-date a communication wave crossing a reference day (section 4.2).

3 Related literature

Price ceilings as focal points. Theory predicts that a binding ceiling can coordinate oligopolists on the administered price when demand is inelastic. The best empirical evidence comes from China's administered gasoline ceilings, which act as focal points producing price uniformity

(Zhang et al. 2020). Legislated gasoline price ceilings have been studied in Canada (Sen et al. 2011), Greece (Gatsios et al. 2026) and Colombia (Quiguanas et al. 2025), generally finding modest price effects and concerns about capture; Spain's transition from ceiling to liberalised pricing is documented by Contín-Pilart et al. (1999). Closest to our counterfactual exercise, Carranza et al. (2015) study price controls in gasoline retail and show their effects depend on market structure. Our setting differs decisively: the cap is *voluntary* and unenforced, so adoption is a firm choice. We observe the diffusion path itself, and we find uniformity is partial — a persistent non-adopter tail coexists with a mass of followers, in line with the dispersion persistence found in regulated Colombian retailing (Quiguanas et al. 2025).

Crude-oil pass-through. A large literature documents asymmetric pass-through — "rockets and feathers" — from crude to retail gasoline (Borenstein, Cameron, et al. 1997; M. Tappata 2009), driven by consumer search (Lewis 2011) and stronger where local market power is high (Deltas 2008; Verlinda 2008); see also Bettendorf et al. (2003) for the Dutch market and Atil et al. (2013) for nonlinearities. Albulescu and Mutascu (2021) document strong fuel-price comovement across France, Germany and Italy, consistent with a common crude-oil driver. This literature establishes both that retail fuel prices track crude with distributed lags — the identifying assumption behind our counterfactual — and that pass-through is state-dependent, which is why we let the data choose the lag structure rather than impose one.

Retail fuel price dynamics. Retail fuel prices are strategic complements with locally heterogeneous adjustment — the mechanism our hazard estimates quantify. Sticky rather than cycling pricing is the norm in several markets (Noel 2007a; Noel 2007b; Haas 2024), consistent with what we observe at the cap.

Consumer search and price dispersion. Equilibrium dispersion arises even with identical agents (Reinganum 1979; Stiglitz and Salop 1982); gasoline applications include Chandra and M. E. Tappata (2011) and Noel (2018). Pennerstorfer et al. (2020) show dispersion is non-monotone in information. Our bimodal post-cap distribution — a mass at the cap plus a non-adopter tail — is a stark departure from these smooth-dispersion benchmarks.

Strategic complementarities and leadership. Sequential "domino" pricing is documented in gasoline markets by Atkinson et al. (2009). Second-mover advantage is generic in Bertrand settings (Amir and Stepanova 2008); direct estimates of price-setting complementarities are provided by Amiti et al. (2020) and, in housing, Guren (2018) — momentum arising from complementarities is precisely the adoption pattern we measure. Network oligopoly structure shapes markups (Pellegrino 2025) and vertical integration affects rivals' costs (Hastings and Gilbert 2005; Borenstein and Shepard 2002).

The Italian fuel market. Price-transparency shocks on Italian motorways cut nearby prices by about 3% (Rossi and Chintagunta 2015); spatial interaction and territorial factors shape Italian retail fuel pricing (Bergantino et al. 2020; Alderighi and Baudino 2015); and non-adoption of price cuts has distributional stakes (Mattioli et al. 2023). Price-cap incentive theory originates in the utility-regulation literature (Drazen and Masson 1993; Cabral and Riordan 1991).

Contribution. We provide the first station-level, communication-based measurement of a *voluntary* cap's diffusion, showing that a unilateral cap triggers competitor matching along a spatial gradient; we quantify the roles of pre-cap price position, brand, chain size and local spillovers in the adoption decision; we build a tax-immune oil-based counterfactual that quantifies the cap's consumer-price effect over a window in which crude rose by three quarters; and we document a reference-day sampling bias of 3.5–4.1 c/l for official statistics when a communication wave

crosses a measurement date, together with the cap's contribution to headline month-on-month inflation.

4 Data

4.1 Sources

We combine six datasets:

- **MIMIT price communications** (daily extracts in our MongoDB cache): every price communication by every Italian fuel station, with station identifier, fuel, price, and communication timestamp (`dtComu`).
- **MIMIT station registry** (*anagrafica*): Gestore (operator), Bandiera (brand), type, municipality and coordinates.
- **ISTAT 2021 census population grid**: resident population in 1 km^2 cells (EPSG:3035).
- **ISTAT administrative boundaries** (2026): province and region polygons.
- **Brent crude and USD/EUR exchange rate**: daily spot Brent (US$/bbl, EIA series via FRED `DCOILBRENTEU`) and the ECB euro reference rate (US$ per €, via FRED `DEXUSEU`). We convert to euro per barrel ($B^e = B^{\$}/e_{\$/e}$) and forward-fill weekends and TARGET holidays onto the daily fuel-price calendar (both series publish on all 88 business days of the window; the conversion moves the euro appreciation in our favour by only about 2% over four months).
- **Excise calendar**: statutory rates per fuel with application dates, from the official decrees recorded in our tax cache (section 2); VAT at 22% throughout.

Our window runs from 1 July to 31 October 2026 and covers *stradale* stations only (motorway stations are a separate market). The adoption analyses of sections 6 and 7 use communications through 12 October (censoring date, see below); the price, counterfactual and inflation analyses use effective prices through 31 October. The station registry covers 39,423 stradale station–fuel units (19,746 petrol, 19,677 diesel); the daily effective-price panel averages about 19,800 petrol and 19,700 diesel stations per day, 98% of which report essentially every day (the balanced-sample robustness of section 9 uses this 98%).

4.2 From communications to effective prices

Daily extracts snapshot the market before each day's communication wave (extracts close at 8am; the bulk of communications arrives between 6:20 and 8:00). A station's "price on day D" read off the extract may therefore reflect a communication made days earlier. We instead define the effective price on day D as the last communication with `dtComu` $\leq D$, propagated forward. A communication made on day D prices day D.

Two consequences. First, adoption dates are exact: a station counts as adopted on the first post-cap day its effective price is at or below the cap. Second, we can measure the extract's *snapshot bias*. On 1 October 2026 — an ISTAT reference day — zero same-day communications appear in the day's extract: the recorded national average overstates the true day-1 average by 3.47 c/l for petrol and 4.06 c/l for diesel (9,192 and 9,221 stations communicated that day, nearly all after the cutoff; on the stradale estimation sample used in section 10 the same bias is 4.2 and 4.9 c/l). Official statistics reading prices off extracts on reference days are subject to this bias whenever a wave crosses a sampling date.

4.3 From pump prices to net prices

The pump price decomposes as $P^{gross} = (P^{net} + \tau) \times (1 + \text{VAT})$, so the net price is $P^{net} = P^{gross}/1.22 - \tau(\text{fuel}, t)$, with the excise τ following the statutory calendar of section 2. Net prices are the object of the counterfactual analysis: they remove every excise and VAT intervention by construction, isolating what the market sets. Around excise steps the net price is noisy for a few days — stations pass the tax change at different speeds, so the net price temporarily absorbs margin — transitory blips that the net-price regression of section 9 simply leaves in the error, since a tax-stripped target should not respond to tax dates.

4.4 Variables

- *Adoption*: effective price $\leq$ cap (2.00/2.20 €/l) on day $t \geq$ 2026-09-28. Time to adoption T_i measured in days from 28 September; stations never adopting are censored at 12 October.
- *Distance to nearest station* (`nn_dist_km`) and *nearest at-cap station* (`cap_dist_km`): great-circle distances in EPSG:3035 computed daily over all priced stations of the same fuel (KD-tree).
- *Local population*: 2021 census population of the station's containing 1-km cell (`pop_density_cell`).
- *Chain size*: number of stations nationwide under the same Gestore (`n_stations_gestore`).
- *Pre-cap gap*: mean effective price 21–27 September minus the fuel-specific cap (`pre_gap_vs_cap`).
- *National daily price series*: the cross-sectional mean of effective prices per fuel and day (self-service, stradale), separately in gross and net terms; the basis for section 9 and section 10.
- *Brand* (`bandiera`) and *region* fixed effects.

Coordinates are harmonised to WGS84; 99.8% of stations are assigned to a province (2 broken-coordinate stations dropped; 53 unassigned stations kept with missing province).

5 Descriptive evidence

Adoption was fast and front-loaded. Figure 1 plots the exact daily share of station–fuel units priced at or below the cap, one panel per fuel. On the cap day itself 21% priced at or below the threshold; the share then rose to 51% by day 4. The petrol and diesel paths track each other almost one-for-one through 5 October (54.0% and 55.1%). On 6 October the government removed the remaining 5 c/l excise discount on diesel, and we raise the diesel cap threshold accordingly (by $5 \times 1.22 = 6.1$ c/l including VAT; the dotted line in the lower panel). The diesel share jumps 11 percentage points overnight, from 55.1% to 66.3%, while petrol is unchanged — this discontinuity is the tax adjustment at work, not diffusion.

The jump reflects genuine repricing rather than a communication or coding artefact. Between 5 and 6 October, 42% of tracked diesel stations raised their price by 4–8 c/l (mean change +2.3 c/l), and 66% of the stations priced at or below 2.20 €/l on 5 October reappeared in the (2.20, 2.261] €/l band the next day. Of the diesel units counted compliant on 6 October in that band, 90% carry a price communication dated 6 October itself (only 10% are carried forward from earlier dtComu), so operators actively re-communicated prices just under the new threshold rather than the threshold move mechanically reclassifying stale prices. By 12 October, 62.5% of petrol and 78.0% of diesel units price at or below the (fuel- and date-adjusted) cap.

Brand gradients are stark. By 12 October, the share of station–fuel units at or below the (fuel- and date-adjusted) cap was: Agip Eni 98.1%, Q8 93.7%, Tamoil 52.4%, Api-Ip 31.6%, Esso 35.2%, Pompe Bianche (independents) 27.3%. Mean prices order identically (fig. 2): the market

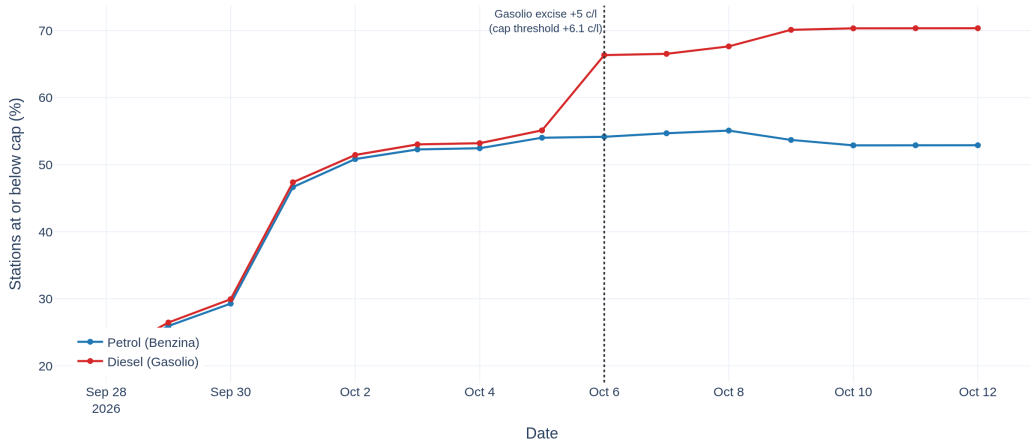


Figure 1: Share of stations priced at or below the cap, by day and fuel. Exact daily shares from the dtComu-effective panel; no model, no uncertainty bands. The dotted line in the lower (diesel) panel marks the excise increase of 6 October (cap threshold +6.1 c/l); petrol is unaffected, so its panel has no line.

split into a cap-following tier and an above-cap tier, with Agip Eni locking at 1.99 €/l from the cap day while Q8 followed only with its 1 October wave.

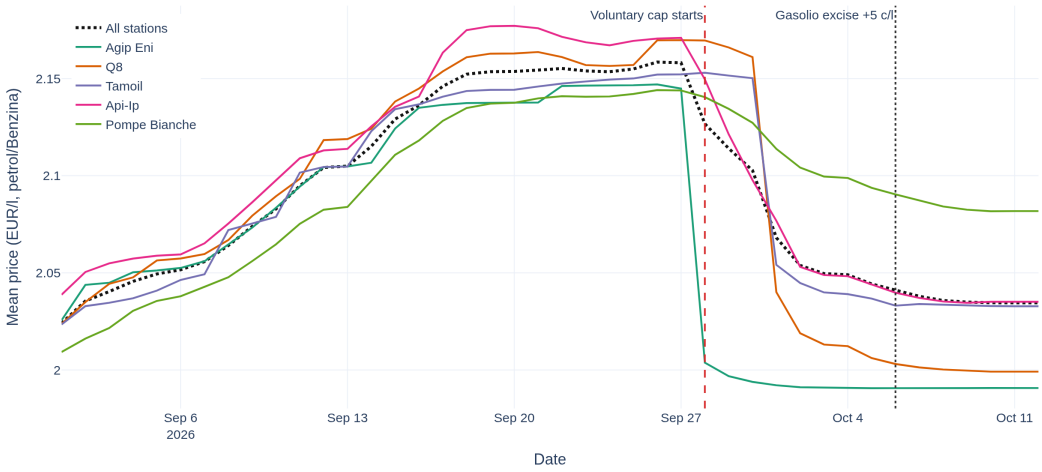


Figure 2: Daily mean pump price (petrol), 1 September – 6 October 2026: all stations (black dotted) and the five largest brands. Dashed red line: voluntary cap starts (28 Sep); dotted line: diesel excise increase (6 Oct).

Geographically, adoption was highest around Rome and the central-northern coastal corridors and lowest in the Po-valley fringes and parts of the South (fig. 3).

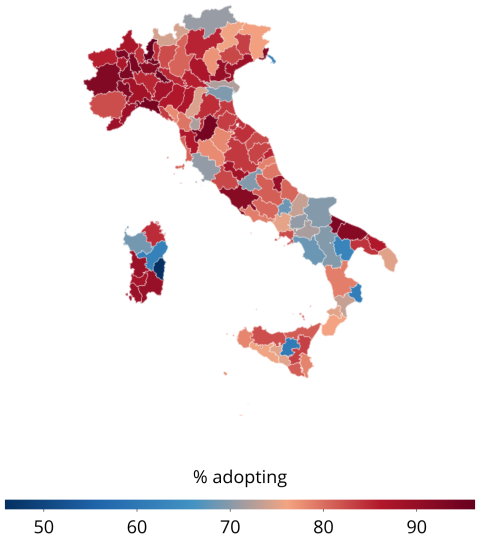


Figure 3: Share of stations adopting the cap by province (all stations; a station counts as adopting if it priced at or below the tax-adjusted cap on at least one day, 28 Sep – 12 Oct 2026; each station counted once regardless of fuel).

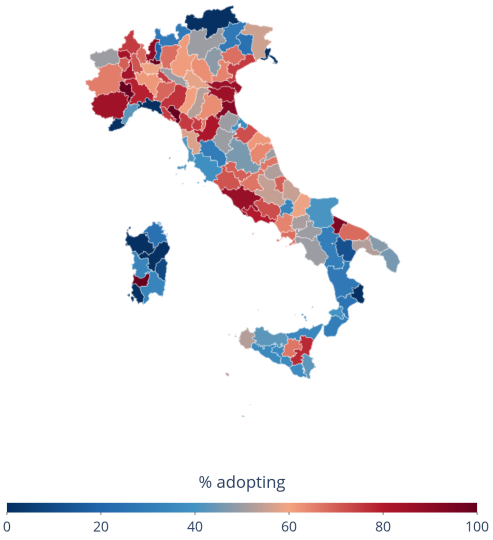


Figure 4: Share of Pompe Bianche (independent) stations adopting the cap, by province.

The overall map (fig. 3) shows a clear geography: adoption was highest around Rome, along the Tyrrhenian corridor and in parts of the North-West (up to 69% of units), and lowest in Alpine provinces and parts of the South and the islands (down to 31%). The range is wide — a 38-percentage-point gap between the extremes — and the north–south gradient is not monotone: some southern provinces (Abruzzo's corridor, coastal Puglia) match northern adoption levels. The independents' map is strikingly different (fig. 4): independent adoption rates are low almost everywhere (national mean 23%, and exactly zero in the darkest provinces) with a sharp pocket of convergence along the central Adriatic axis — from Umbria through the Marche – Abruzzo corridor — where independents followed the leader's network, and near-zero adoption in the Alpine arc and the islands. Independents responded to local pressure, not to the national announcement.

6 Who adopts? Survival analysis

We estimate Cox proportional-hazards models (Cox 1972) of time to adoption. Duration is days from 28 September to first at-cap day; never-adopters are right-censored at 12 October, and the diesel threshold is tax-adjusted from 6 October. Because brand differences are so stark in the raw data — and because the leader's own stations adopted almost mechanically by construction — we estimate the model on three samples: the *universe* of station–fuel units; the market *excluding* Agip Eni and Q8 (the two brands that announced at-cap prices in the first wave); and *Pompe Bianche only* (independent stations). Standard errors are clustered by Gestore (operator). Chain size counts stations nationwide under the same Gestore.

Table 1: Cox duration-model estimates by sample (coefficients; hazard ratios = $e^{\hat{\beta}}$)

	(1) Universe	(2) Excl. Agip Eni + Q8	(3) Pompe Bianche
Nearest station (km)	-0.0994***	-0.0790***	-0.1306***
Nearest day-0 adopter (km)	-0.0251***	-0.0516***	-0.0623***
Population per 1-km cell	0.0000**	0.0000**	-0.0000
Pre-cap gap (€/l)	-4.2710***	-6.5040***	-5.4960***
Chain size (Gestore, stations)	0.0007***	0.0010***	0.0071***
Diesel	0.2522***	0.4398***	0.6421***
N	39,423	26,714	5,198
Events	27,691	15,304	2,112

Gestore-clustered robust standard errors. All models include a diesel dummy. The diesel cap threshold is adjusted for the excise change of 6 October 2026 (see section 4).

Table 1 reports coefficients (hazard ratios are $e^{\hat{\beta}}$). Three results are robust across samples. First, the **pre-cap gap dominates**: its coefficient grows (in magnitude) from −4.2 in the universe to −6.5 excluding the leader brands and −5.4 for independents — stations already pricing near the cap adopted almost mechanically, and this remains the single strongest predictor once leader stations are removed. Second, **spatial spillovers strengthen when the leader brands are removed**: the nearest-adopter distance coefficient roughly doubles from −0.025 (universe) to −0.052 (excl. Agip Eni/Q8) and −0.062 (Pompe Bianche). This is exactly the pattern strategic-complementarity models predict: leader-brand rows absorbed the spatial variation, and once they are excluded, the neighbourhood diffusion effect emerges clearly. Independents respond to what nearby stations do, not to the cap announcement itself. Third, **chain size matters most for independents** (0.0016 versus 0.0001): small local chains (gestori) coordinated their response, while brand-level headquarters decided for the organised brands.

Robustness. A Weibull AFT model gives the same signs and significance per sample; gestore-clustered standard errors leave all conclusions unchanged; the PH assumption is examined in appendix C and the reported coefficients should be read as window-averaged effects.

7 Market dynamics: panel event study and adoption probabilities

7.1 Event study

We estimate station fixed-effects models of the daily at-cap indicator over the event window $[-7, +7]$ around the cap:

$$\text{at_cap}_{it} = \sum_{k \neq -1} \beta_k \, \mathbf{1}[t - T_0 = k] + \alpha_i + \varepsilon_{it}, \tag{1}$$

with day −1 as reference and gestore-clustered errors. Because fig. 1 already shows the raw adoption path — the data are exhaustive, so the shares themselves carry no sampling uncertainty — the event study serves only as a robustness check that the pattern is not an artefact of composition. Its coefficients are reported in appendix A and mirror fig. 1: no pre-cap movement, a jump at day 0, and a plateau from day 5.

To trace diffusion, we split station–fuel units into terciles by their distance to the nearest station that adopted on day 0 (cut points: 1.08 and 3.15 km) and plot the exact daily at-cap share of each tercile in fig. 5. The three curves never cross. At every point in time, stations near a day-0 adopter adopt earlier and more often: on 5 October, 62.7% of the near tercile is at the cap versus 44.3% of the far tercile — an 18-percentage-point gradient — and the far tercile never catches up over our window. Note that the far tercile's curve is not flat either: it rises throughout, so diffusion eventually reached remote stations too, but with a lag of several days. This ordering is exactly what local strategic complementarities predict, and it is invisible in the aggregate series of fig. 1.

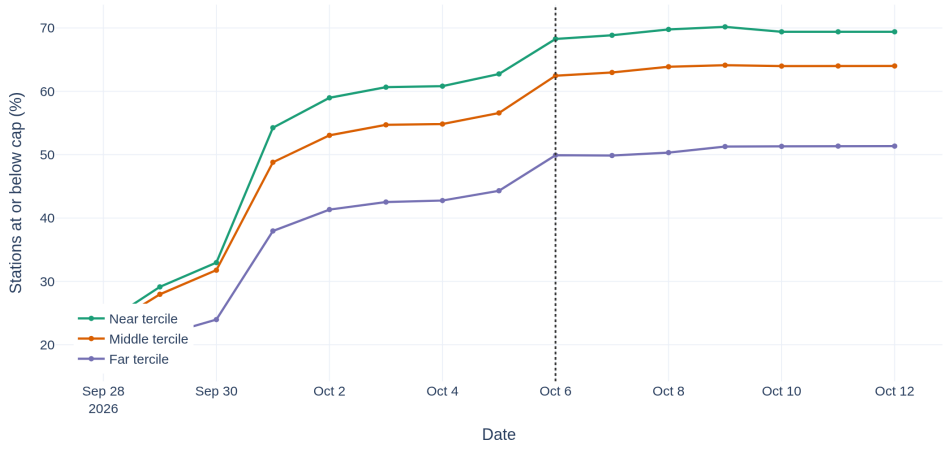


Figure 5: Stations at or below the cap by distance to the nearest day-0 adopter (terciles; exact daily shares). Cut points: 1.08 and 3.15 km.

7.2 Who adopts? Adoption probabilities

The survival analysis of section 6 conditions on time; here we ask the complementary question: given the full covariate set of section 4.4, *which* stations ended up adopting at all? We estimate linear probability models (OLS with gestore-clustered standard errors)

$$\text{adopted}_i = \mathbf{x}_i'\gamma + \mu_{b(i)} + \delta_{r(i)} + u_i, \tag{2}$$

where $\mathbf{x}_i$ collects the continuous and fuel covariates, μ_b are brand fixed effects (within-sample ≥100-observation rule) and δ_r region fixed effects. Table 2 reports the core coefficients for the same three samples as table 1; the full brand and region coefficients appear in tables 6 and 7.

The LPM confirms and sharpens the survival results. The pre-cap gap is by far the strongest predictor in every sample: −1.31 probability points per cent/litre of pre-cap excess in the universe sample, −1.56 excluding the leader brands, −1.05 for independents. Distance to the nearest day-0 adopter is negative and significant in all three samples (−0.010 universe, −0.014 excluding the leader brands, −0.010 for Pompe Bianche), while the distance to the nearest station (competitor) is smaller and only marginally significant for independents (−0.018, $p = 0.009$): proximity to *adopters* predicts adoption at least as strongly as proximity to competitors, and for independents

Table 2: Adoption probability (LPM) by sample — core covariates

	(1) Universe	(2) Excl. Agip Eni + Q8	(3) Pompe Bianche
Intercept	1.0611***	0.4366***	0.3930***
Nearest station (km)	-0.0161***	-0.0160***	-0.0175**
Nearest day-0 adopter (km)	-0.0106***	-0.0145***	-0.0116***
Population per 1-km cell	-0.0000	0.0000	0.0000
Pre-cap gap (€/l)	-1.3068***	-1.5584***	-1.0522***
Chain size (Gestore, stations)	0.0001*	0.0003**	0.0016***
Diesel	0.1449***	0.2062***	0.2295***
Brand & region FE	Yes	Yes	Region only
N	39338	26662	5180
Adj. R^2	0.3387	0.2496	0.2488

Dependent variable: adopted (=1 if the station–fuel unit priced at or below the cap on any day 28 Sep – 7 Oct 2026; the diesel cap is tax-adjusted from 6 Oct). OLS with gestore-clustered standard errors. Brand reference: Agip Eni in columns (1), Ala in column (2); region reference: Abruzzo. Full brand and region coefficients in tables 6 and 7. In column (3) the sample is single-brand (Pompe Bianche), so brand FE are undefined.

the two effects are of similar magnitude once measurement noise is accounted for. Chain size is positive and significant in the universe and Pompe Bianche samples (insignificant once the leader brands are removed, whose networks coordinate centrally) and roughly seventeen times larger for Pompe Bianche, echoing the gestore-coordination reading of section 6. Population density is economically negligible once brand and region are controlled. Fit is moderate (adjusted $R^2 = 0.25$–0.34), as expected for individual adoption decisions; diagnostics are in appendix C.

8 Dispersion and competitiveness

Province-level dispersion. We compute, for each province × fuel × phase (pre: 21–27 September; post: 28 September – 12 October), the daily within-province standard deviation of effective prices, and compare phases. The SD rose in **214 of the 220 province–fuel cells** (97.3%): on average from 51 to 87 c/l for petrol and from 65 to 90 c/l for diesel — a doubling. Figures 19 and 20 show the distribution of the shift. Throughout the paper, maps use a blue–red scale: *blue marks low values and red marks high values.* The province-level standard deviation rises from a median of 50 c/l pre-cap to 90 c/l post-cap, and the whole distribution shifts right (fig. 6, top panel): 214 of the 220 province–fuel cells see their SD increase. Figures 19 and 20 in appendix D show the geography: pre-cap the map is predominantly light blue (typical SDs 28–150 c/l); post-cap red spreads along the Alps and the Apennines — the provinces where dispersion widened most are those where the adopter share was intermediate, mixing cap-priced with above-cap stations. Provincial SDs after the cap run 71–235 c/l, and no province returns to its pre-cap level. The cap did not compress the distribution — it split it. A mass of adopters sits exactly at the threshold while a non-adopter tail holds prices above it, producing a bimodal distribution unlike the smooth equilibrium-dispersion benchmarks (Reinganum 1979; Stiglitz and Salop 1982).

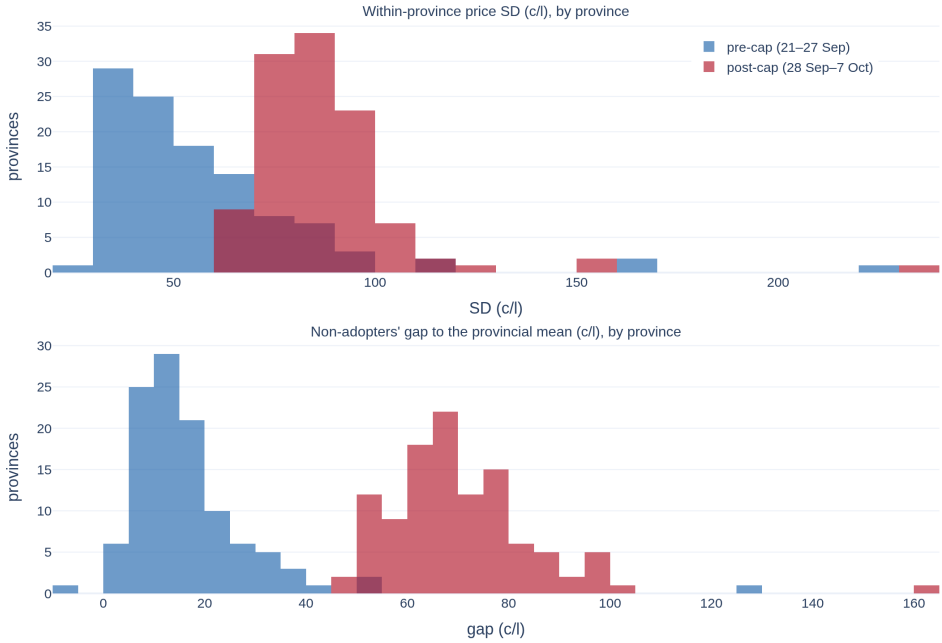


Figure 6: Province-level distributions before (blue) and after (red) the cap. Top: within-province standard deviation of effective prices. Bottom: non-adopters' mean gap to the provincial daily mean.

The consumer reading is correspondingly uneven: where adoption was high, prices fell to the cap for most of the market; where the non-adopter tail dominated, prices barely moved. Given that adoption tracked population density and chain presence, the gains concentrate in urban, organised-firm corridors — with distributional implications for car-dependent rural areas (Mattioli et al. 2023).

9 Counterfactual prices: what would the market have done?

The adoption evidence says who followed the leader; it does not say what the cap was worth. Over our window Brent crude in euro rose from an index of 100 (1 July) to 173 on the cap date and 177 at the end of October — a supply-side impulse that, absent the cap, would have pushed October pump prices up. Measuring the cap therefore requires a counterfactual net-price path: what the market's net margins would have been had the pre-cap relationship with oil continued.

9.1 Net prices tracked oil — until the cap

Figure 7 normalises the three relevant series — Brent in euro per barrel, the national net price of petrol and the national net price of diesel — to 100 on 1 July 2026, the first day of our window. Two facts stand out. First, before the cap the three series move together: over 1 July–27 September, Brent rises 67% and the two net prices rise 28% and 44% — co-movement, with the net prices responding less than proportionally and diesel more than petrol, consistent with partial and lagged pass-through (Borenstein, Cameron, et al. 1997; M. Tappata 2009). Second, from 28 September the net prices stop following oil: petrol's index falls from 125 to 114 and diesel's from 140 to 130 by 31 October, while Brent's climbs to 177. The visual break is the cap at work on the market's margin.

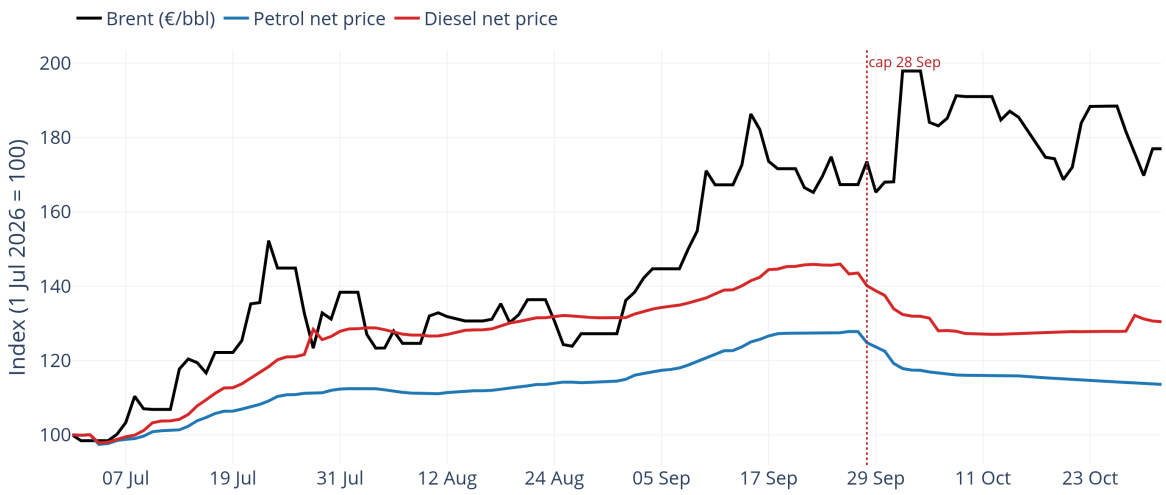


Figure 7: Brent crude (€/bbl) and national net prices (petrol, diesel), all indexed to 100 on 1 July 2026. The red dotted line marks the voluntary cap (28 September). Before the cap the three series co-move; after it, the net prices decouple from oil.

9.2 An ARDL model of net prices

We model the daily national net price of each fuel as an autoregressive distributed-lag (ARDL) process in logs,

$$\ln P_t^{net} = \alpha + \sum_{i=1}^{p} \phi_i \ln P_{t-i}^{net} + \sum_{j=0}^{q} \theta_j \ln B_{t-j}^{e} + \varepsilon_t, \tag{3}$$

where B^e is Brent in €/bbl. Because the dependent variable is net of excise and VAT by construction, the tax calendar is already stripped from the target and no tax term appears in the equation: statutory tax changes should not move a net price, and the transitory blips they do leave in the measured series — stations pass a tax change at different speeds — stay in the error, where they belong. The ARDL form (Pesaran et al. 2001) lets the data pick both the speed of pass-through and the long-run level relationship; we search $(p, q) \in [0, 10]^2$ by AIC and BIC on the pre-cap window, 1 July–27 September (84–88 daily observations after lags). For diesel both criteria agree on $(1, 1)$; for petrol AIC selects $(4, 5)$ and BIC the more parsimonious $(5, 0)$.

Table 3 reports the specification and fit. The models track the net prices closely ($R^2 = 0.999$ petrol, 0.995 diesel), and the implied *long-run pass-through elasticity* of net prices to Brent in euro — $\sum_j \theta_j / (1 - \sum_i \phi_i)$ — is 0.44 for petrol and 0.63 for diesel: a 10% rise in the euro oil price raises the net price by 4.4–6.3% once lags play out. The adjustment half-life is about 25 days for petrol and 9 days for diesel. Diagnostics (appendix C: Breusch–Godfrey, RESET, Newey–West standard errors) are discussed there; in brief, petrol's Breusch–Godfrey statistic is borderline at short lags (read inference with the reported Newey–West errors) and the diesel RESET rejects ($p \approx 0.008$, driven by the two large July excise transitions), acceptable for a forecasting device estimated on 84–88 observations.

Placebo backtest. Before trusting a projection, we check that the design can forecast a period it was not fitted on. Refitting to 13 September and projecting 14–27 September — still pre-cap, so the truth is observable — the recursive projection attains a root-mean-square error of 1.80 c/l (petrol) and 5.03 c/l (diesel) on the net price, against 3.69 and 5.07 c/l for a no-change random walk, with biases of +1.8 and −4.1 c/l. The ARDL halves the benchmark error for petrol; for diesel it merely ties the random walk, its edge eroded by the two excise steps inside the backtest window (18 and 26 September — excluding those transition days the RMSE falls to 4.05 c/l) and

Table 3: ARDL net-price models: specification and out-of-sample fit

	Order (p, q) (AIC / BIC)	Long-run elasticity (net price vs Brent €/bbl)	RMSE ARDL (c/l)	RMSE random walk (c/l)	N
Petrol	(4,5) / (5,0)	0.437	1.80	3.69	84
Diesel	(1,1)	0.630	5.03	5.07	88

Dependent variable: log national average net price (self-service, stradale), daily, 1 Jul–27 Sep 2026; regressor: log Brent in €/bbl (ECB reference rate), lags as shown. The target is net of excise and VAT by construction, so no tax dummy enters the regression. Lag orders selected by AIC and by BIC over $(p, q) \in [0, 10]^2$ (the AIC selection is used throughout; where the two differ both are shown). Backtest: refit to 13 Sep, projected 14–27 Sep (pre-cap) recursively; RMSE on the net price in c/l against a no-change random walk. See section 9.

by a modest upward drift in the projection. The bias signs also bound the counterfactual: petrol's device under-predicts the net price, making that gap conservative, while diesel's over-prediction would, if it carried into October, shave roughly 4 c/l off a 27 c/l gap.

9.3 The counterfactual pump price

From 28 September we project eq. (3) recursively: the model's own projected net prices feed its lags, while Brent follows the actual path — actual post-cap net prices are never used, since the cap contaminated them. The counterfactual pump price is then $\widehat{P}_t^{gross} = (\widehat{P}_t^{net} + \tau_t) \times 1.22$, with the *actual* excise calendar τ_t: both the observed and the counterfactual market face the same taxes, so the comparison isolates the cap, not tax policy. Uncertainty bands come from a residual bootstrap (2,000 draws, 95% intervals).

Figure 8 shows the result. The counterfactual petrol price starts where the market started (2.16 vs 2.13 €/l on 28 September, a gap of 3.2 c/l) and climbs with oil while the actual price falls to the cap: by 31 October the gap is 20.7 c/l. Diesel's counterfactual climbs from 2.38 to 2.56 €/l — a 31.8 c/l gap at the end — while the actual price stays pinned near the (tax-adjusted) cap. Averaged over the 34 post-cap days, the cap saved consumers 16.5 c/l per litre of petrol and 27.5 c/l per litre of diesel relative to the oil-following counterfactual. The bands widen slowly (±3.3 c/l for petrol and ±7.5 c/l for diesel at the 34-day horizon), so the gap remains a multiple of its uncertainty from mid-October on.

Robustness. Four checks. First, for petrol BIC selects the more parsimonious ARDL$(5, 0)$; its long-run elasticity (0.46) sits within 0.02 of the AIC selection's, and for diesel both criteria agree on $(1, 1)$ (table 3). Second, restricting the sample to the 98% of stations that report essentially every day (a balanced panel) moves the national daily means by fractions of a cent — composition is not driving the gap. Third, using the naive 8am-cut extract averages instead of effective prices would add the snapshot bias of section 4.2 to the actual series on reference days (4–5 c/l *against* the cap), making the counterfactual gap larger, not smaller. Fourth, the result is not an artefact of the euro: the USD/€ rate moved less than 2% over the window, and converting Brent at any pre-window average rate changes the October gaps by well under 1 c/l. The main caveat is sample length: 85–88 daily observations support a parsimonious ARDL but not a full cointegration analysis, and the projection asks the pre-cap dynamic to persist for 34 days — the backtest suggests this is a reasonable, not a costless, assumption.

10 From pump prices to measured inflation

ISTAT measures fuel prices for the NIC on the 1st, 11th and 21st of each month; the monthly fuel index moves with the average of the three reference days, and motor fuels carry 4.5219% of the 2026 basket (definitive weights). Table 4 compares the realised September-to-October

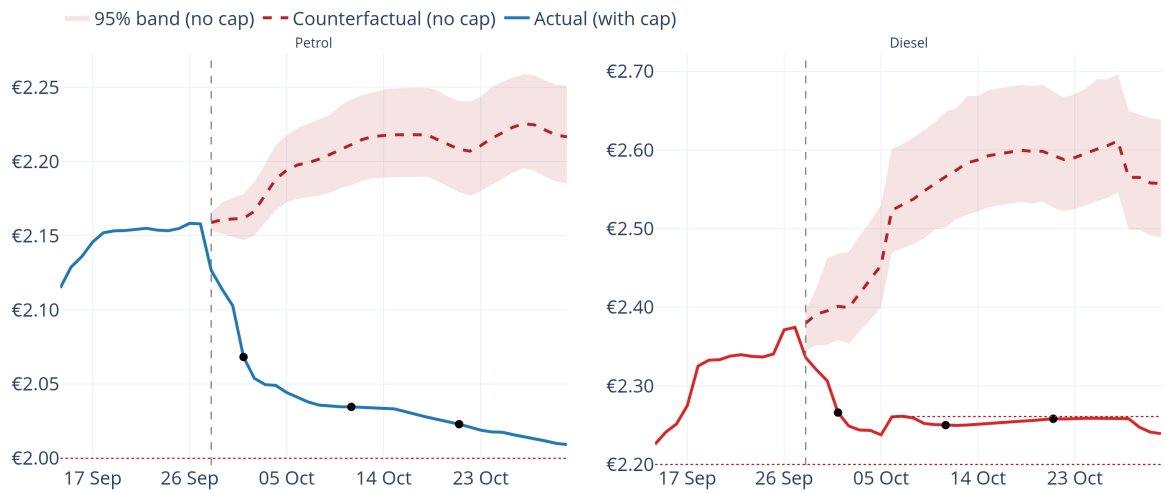


Figure 8: Actual pump price (with cap) vs counterfactual (no cap, ARDL projection on the actual Brent path, 95% bootstrap band), 14 September – 31 October 2026. Dotted red lines: the tax-adjusted cap levels (diesel steps to 2.261 €/l on 6 October). Grey dashed line: cap start. Black dots: ISTAT reference days.

month-on-month change of the national average pump price — averaged over the reference days, exactly the object the index tracks — with the counterfactual of section 9 on the same days. Pre-cap reference days (September) have no counterfactual by construction: there the no-cap path *is* the observed path.

Table 4: Month-on-month inflation on ISTAT reference days (1st, 11th, 21st): actual vs counterfactual

	MoM change, Sep vs Oct avg (%)			Contribution to headline MoM CPI (pp)		
	Actual	No cap	Difference	Actual	No cap	Difference
Petrol	−2.34	+4.92	−7.26	−0.05	+0.11	−0.16
Diesel	+1.58	+13.37	−11.79	+0.04	+0.30	−0.27
Motor fuels (combined)	−0.38	+9.15	−9.52	−0.02	+0.41	−0.43

Monthly averages over the three ISTAT reference days (1, 11, 21 September vs 1, 11, 21 October 2026), national mean pump prices (self-service, stradale). Pre-cap reference days have no counterfactual (the no-cap path is the observed path). CPI contributions use the NIC 2026 definitive weight of ECOICOP 07.2.2 "Carburanti e lubrificanti" (4.5219% of the basket), split evenly between petrol and diesel.

The result is stark. With the cap, the October reference-day average was 2.042 €/l for petrol and 2.258 €/l for diesel, month-on-month changes of −2.34% and +1.58%: petrol *fell* into an oil-price surge. Without the cap, the counterfactual implies 2.194 and 2.520 €/l — +4.92% and +13.37% month-on-month. The cap therefore changed the fuels' month-on-month rate by −7.26 percentage points (petrol) and −11.79 (diesel), or −9.52 pp for motor fuels combined. Note that both series face the same tax calendar: the diesel excise steps of 6 October fall on the actual and the counterfactual alike, so the difference is conduct, not fiscal policy.

Weighted by the NIC basket, motor fuels contributed −0.02 pp to headline month-on-month October inflation with the cap, against +0.41 pp in the counterfactual: the voluntary cap lowered October's measured month-on-month inflation by roughly **0.43** percentage points. For a measure framed against "speculation" during an oil surge, this is the cleanest statement of what it bought: about a third of the counterfactual month's fuel-price inflation was absorbed by the market's own margin compression rather than passed to pumps.

A measurement caveat ties back to section 4.2: the *recorded* (8am-cut) averages on 1 October overstate the effective ones by 4.2 c/l (petrol) and 4.9 c/l (diesel) on our sample, because the adoption wave communicated after 8am was mis-dated. Official October fuel inflation thus mixes the cap effect with a data artifact in the first half of the month; effective-price reconstruction, as used here, is the robust measurement.

11 Discussion

What did the cap buy, and at whose cost? Our counterfactual prices the cap's consumer benefit at 16.5 c/l (petrol) and 27.5 c/l (diesel) per litre over the first post-cap month — 0.43 pp of October headline inflation. Unlike an excise cut, which lowers prices for everyone at direct fiscal cost, the voluntary cap operates on margins: the leader gave up its own premium and invited rivals to compress theirs. The cost side is invisible in our data — who absorbed the cut (refining margins, retail margins, or the leader's brand premium) is a question for company accounts — but the measurement is not: the benefit accrues wherever adoption happened.

The cap and the tax lever interact. The 2026 excise calendar moved seven times, with steps up to 14 c/l — the same instrument the cap was defending consumers with. On 6 October the diesel cap itself rose by 6.1 c/l because the state removed a discount; fig. 1 shows the compliance share jumping 11 pp overnight as operators repriced just under the new threshold. An anti-inflation commitment on net prices would be immune to this interference; a cap defined on gross prices is hostage to every tax move.

A voluntary cap polarises. The adoption analysis (section 8) and the price levels (fig. 8) tell one story from two angles: the cap compressed the follower mass onto a ceiling and let a non-adopter tail drift upward with oil — province-level dispersion doubled, and non-adopters ended October 20–31 c/l above where a cap-follower stood. The consumer benefit of a voluntary cap is thus rationed by geography and brand, an equity wrinkle for a policy advertised as protecting all drivers (Mattioli et al. 2023).

Sunset effects. The cap was announced for two months. Our window ends on 31 October, roughly a third of the way into that horizon; strategic behaviour around expiry — front-loading purchases, or a coordinated exit — is outside these data and is the natural next measurement.

12 Conclusions

A voluntary, temporary price cap announced by the Italian market leader achieved partial market-wide matching within days — by 12 October 62.5% of petrol and 78.0% of diesel station–fuel units priced at or below the (tax-adjusted) threshold. Adoption was primarily a firm decision — but not only: once the leader brands are excluded, local spillovers emerge as a first-order driver, with the adoption hazard falling 7–14% per kilometre from the nearest adopter and independent stations following in dense, organised corridors while ignoring the cap elsewhere. Pre-cap price position dominates everywhere: stations already pricing near the threshold adopted almost mechanically. The cap polarised rather than compressed the market: dispersion widened in almost every province, and non-adopters became relatively more expensive.

The price counterfactual turns these adoption facts into consumer value. An ARDL model of net prices on Brent-in-euro — estimated on pre-cap data only, and validated by a placebo backtest that cuts the random-walk forecast error by half for petrol and ties it for diesel — implies that without the cap, October pump prices would have averaged 16.5 c/l (petrol) and 27.5 c/l (diesel) higher, with the gap reaching 21 and 32 c/l by month end. On ISTAT reference days the

cap changed the month-on-month fuel inflation rate by −7.3 pp (petrol) and −11.8 pp (diesel): realised October fuel-price inflation of −2.3% and +1.6% would have been +4.9% and +13.4% in the oil-following counterfactual. In headline terms, the cap lowered October's month-on-month NIC inflation by roughly 0.43 percentage points.

Three caveats. First, the adoption window is fifteen post-cap days and the counterfactual extends to thirty-four; the two-month cap horizon may produce further convergence or reversal around expiry — strategic pricing near sunset dates is unobservable here. Second, adoption is measured from communications; enforcement (actual pump prices versus communicated ones) is outside our data. Third, the counterfactual assumes the pre-cap oil pass-through dynamic would have persisted; it survives a placebo backtest, alternative lag selections, a balanced panel, and any plausible exchange-rate counterfactual, but it cannot rule out that non-cap conduct would itself have changed after 28 September. Coordinates and brand classifications inherit MIMIT data quality.

For measurement, our snapshot-bias result is a caution for official statistics: reference-day sampling from 8am-cut extracts mis-dated an entire communication wave in October 2026, overstating reference-day averages by 3.5–4.1 c/l. Communication-time reconstruction, as implemented here, is the robust alternative.

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Appendix

A Event-study coefficients

Table 5: Station-FE event-study coefficients (eq. (1))

Event time	$\hat{\beta}_k$	Std. error
Day +0	0.193***	0.037
Day +1	0.243***	0.037
Day +2	0.277***	0.037
Day +3	0.451	0.046
Day +4	0.492	0.044
Day +5	0.507	0.042
Day +6	0.509	0.042
Day +7	0.526	0.040

Dependent variable: at-cap indicator; station fixed effects; gestore-clustered errors; day -1 = 27 September omitted. Pre-cap coefficients are small and insignificant; the jump occurs at $k = 0$ and the series plateaus from $k = 5$.

B Full LPM coefficient tables

Brand and region fixed-effect estimates for the linear probability models of section 7.2 are reported in tables 6 and 7.

C Model diagnostics

This appendix reports diagnostics for the Cox models of table 1, the LPMs of table 2, and the ARDL counterfactual models of table 3.

Table 6: Adoption probability (LPM), universe sample — brand and region fixed effects

	Coefficient
Brand: Ala	-0.6558***
Brand: Api-Ip	-0.3013***
Brand: Aquila	-0.7797***
Brand: Beyfin	-0.5302***
Brand: COIL	-0.1048
Brand: CONAD	-0.0536*
Brand: Costantin	-0.0700
Brand: Ego	-0.5019**
Brand: Energas	-0.0905
Brand: Enerpetroli	-0.4023***
Brand: Esso	-0.3559***
Brand: Europam	-0.2253
Brand: Giap	-0.2804***
Brand: Itala Petroli	-0.4221***
Brand: KEROPETROL	-0.7726***
Brand: Loro	-0.7554***
Brand: Lukoil	-0.4155***
Brand: OTHER	-0.4532***
Brand: Petrol Gamma	-0.6090***
Brand: Pompe Bianche	-0.5788***
Brand: Q8	-0.0826**
Brand: Retitalia	-0.5938***
Brand: San Marco Petroli	-0.8421***
Brand: Shell	-0.6260***
Brand: Sia fuel	-0.3243***
Brand: Tamoil	-0.2361***
Brand: Toil	-0.4290**
Brand: Vega	-0.1534***
Region: Abruzzo	0.0981
Region: Basilicata	0.0821
Region: Calabria	0.0804
Region: Campania	0.1015
Region: Emilia-Romagna	0.0463
Region: Friuli-Venezia Giulia	0.0140
Region: Lazio	0.1207
Region: Liguria	0.0989
Region: Lombardia	0.0848
Region: Marche	0.0982
Region: Molise	0.0463
Region: Piemonte	0.0914
Region: Puglia	0.0947
Region: Sardegna	0.0377
Region: Sicilia	0.1005
Region: Toscana	0.0527
Region: Umbria	0.0977
Region: Veneto	0.0807
Brand: Api-Ip	0.3333***
Brand: Aquila	-0.0987*
Brand: Beyfin	0.1470**
Brand: COIL	0.5446***
Brand: CONAD	0.6063***
Brand: Costantin	0.6043***
Brand: Ego	0.1669
Brand: Energas	0.5320***

Table 7: Adoption probability (LPM), excluding Agip Eni and Q8 — brand and region fixed effects

	Coefficient
Brand: Api-Ip	0.3333***
Brand: Aquila	-0.0987*
Brand: Beyfin	0.1470**
Brand: COIL	0.5446***
Brand: CONAD	0.6063***
Brand: Costantin	0.6043***
Brand: Ego	0.1669
Brand: Energas	0.5320***
Brand: Enerpetroli	0.2335***
Brand: Esso	0.3016***
Brand: Europam	0.4173**
Brand: Giap	0.3598***
Brand: Itala Petroli	0.2381***
Brand: KEROPETROL	-0.1128***
Brand: Loro	-0.0911*
Brand: Lukoil	0.2465***
Brand: OTHER	0.2096***
Brand: Petrol Gamma	0.0394
Brand: Pompe Bianche	0.0770**
Brand: Retitalia	0.0737
Brand: San Marco Petroli	-0.1564***
Brand: Shell	0.0124
Brand: Sia fuel	0.3431***
Brand: Tamoil	0.3795***
Brand: Toil	0.2190
Brand: Vega	0.4753***
Region: Abruzzo	0.0912
Region: Basilicata	0.0580
Region: Calabria	0.0575
Region: Campania	0.1016
Region: Emilia-Romagna	0.0157
Region: Friuli-Venezia Giulia	-0.0624
Region: Lazio	0.1172
Region: Liguria	0.0932
Region: Lombardia	0.0660
Region: Marche	0.0854
Region: Molise	0.0048
Region: Piemonte	0.0738
Region: Puglia	0.0828
Region: Sardegna	-0.0320
Region: Sicilia	0.0717
Region: Toscana	0.0120
Region: Umbria	0.0947
Region: Veneto	0.0520
Region: Abruzzo	0.1797
Region: Basilicata	-0.0908
Region: Calabria	-0.0643
Region: Campania	0.1218
Region: Emilia-Romagna	0.0948
Region: Friuli-Venezia Giulia	-0.0852
Region: Lazio	0.1909
Region: Liguria	0.1052
Region: Lombardia	0.0850

Collinearity. The continuous covariates are close to orthogonal: all pairwise correlations are below 0.62, the largest VIF is 1.6 (nearest-station distance), and the condition number of the covariate correlation matrix is 4.5–5.7 across samples — far below conventional concern thresholds (VIF > 10, condition number > 30). Pre-cap gap and the distance measures thus carry independent variation; coefficients are not fragile to the inclusion of the others.

Proportional hazards (Cox). Scaled Schoenfeld-residual tests (`cox.zph`, KM transform) reject global PH in all samples, driven by *chain size* and, in the universe sample, the *pre-cap gap* (their coefficients attenuate as weaker, further-from-cap stations adopt over the window). The qualitative ranking of covariates is unaffected; the reported coefficients should be read as window-averaged effects.

Functional form (Cox). Martingale-residual correlations with each continuous covariate are $\leq |0.08|$ in all samples — no evidence of misspecified linearity for the distance, density and gap variables at this scale.

LPM specification. Pearson residuals have mean zero by construction and are calibrated in the aggregate (mean predicted adoption equals the observed share to four decimals in every sample). The Ramsey RESET rejects exact linearity in the universe and excl.-Agip/Q8 samples ($p < 0.001$), which is expected: the true response surface is a probability bounded in $[0, 1]$, and the LPM is used for its readable coefficients, not for out-of-range extrapolation. Heteroskedasticity inherent to binary outcomes is handled by the gestore-clustered (Huber–White) standard errors reported throughout.

ARDL diagnostics. Petrol ARDL$(4, 5)$: $R^2 = 0.999$, residual SD 0.24% of the net price; RESET (powers 2–3) does not reject ($p = 0.47$); Breusch–Godfrey is borderline at order 7 ($p = 0.053$) and clean at order 14 ($p = 0.20$), so coefficient inference uses Newey–West (lag 7) standard errors. Diesel ARDL$(1, 1)$: $R^2 = 0.995$, residual SD 0.79%; Breusch–Godfrey is clean ($p = 0.78$ and 0.94); RESET rejects ($p \approx 0.008$), reflecting the two large July excise transitions (4 and 28 July), whose staggered pass-through briefly blips the measured net price that the tax-stripped target leaves in the error. Lag-order grids by AIC and BIC over $(p, q) \in [0, 10]^2$ select $(4, 5)$ and $(5, 0)$ for petrol (both reported in table 3) and $(1, 1)$ for diesel; the ARDL package's `auto_ardl` reproduces the AIC selections.

D Provincial maps by fuel

This appendix collects the province maps behind section 8. All maps use the same blue–red scale as the main text: *blue = low values, red = high values.*

Average provincial price the day before the cap (27 September). Both fuels show the pre-cap north-west–south-east price gradient, petrol around 2.00–2.20 €/l and diesel around 2.20–2.40 €/l.

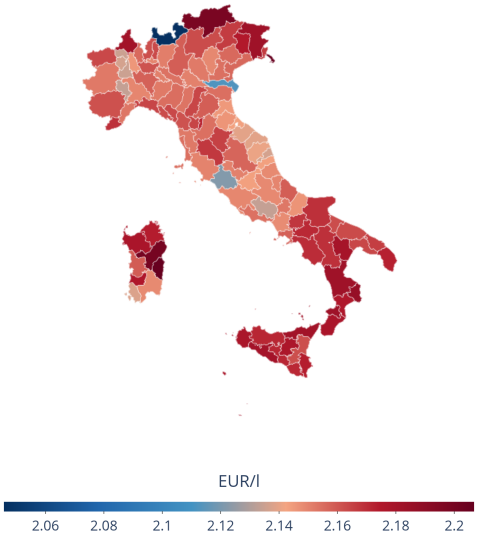


Figure 9: Petrol: mean provincial price, 27 September 2026 (€/l).

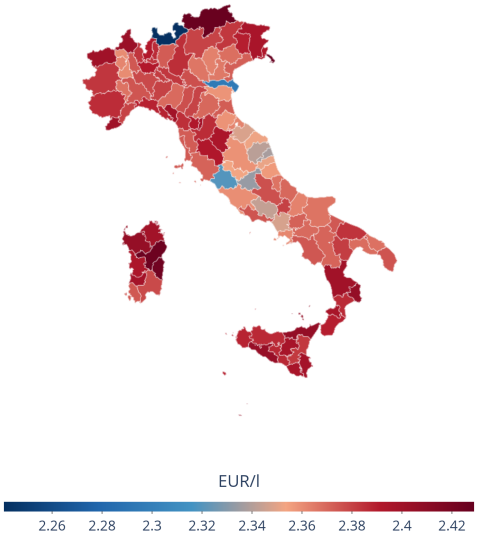


Figure 10: Diesel: mean provincial price, 27 September 2026 (€/l).

Average provincial price 7 days after the cap (5 October). Petrol collapses towards 2.00–2.10 €/l nearly everywhere; diesel converges towards 2.10–2.30 €/l, with the reddest (highest) provinces in the non-adopting fringes.

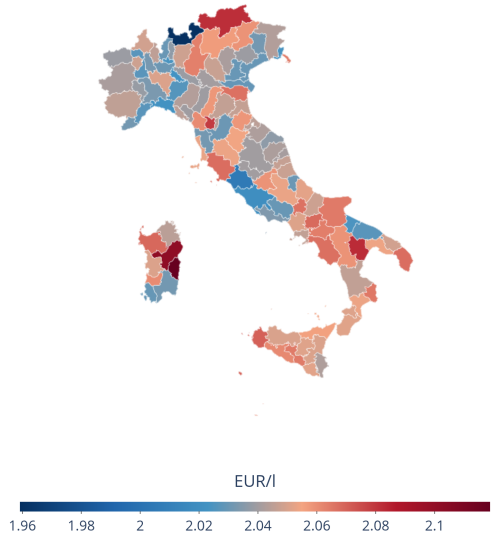


Figure 11: Petrol: mean provincial price, 5 October 2026 (€/l).

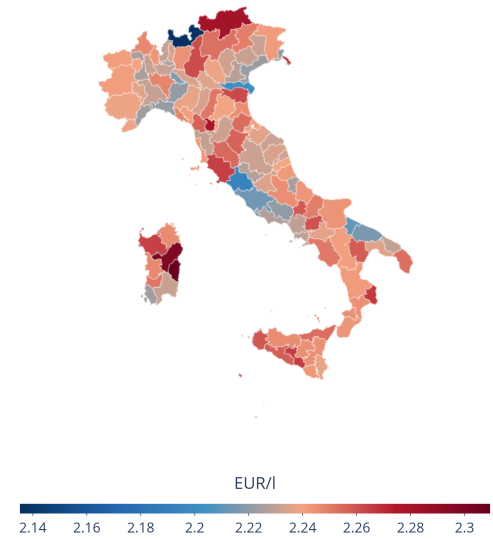


Figure 12: Diesel: mean provincial price, 5 October 2026 (€/l).

Price gap between future non-adopters and adopters before the cap (21–27 September). Even before the cap, stations that would never adopt priced above those that would, by up to 200 c/l (petrol) and 198 c/l (diesel) in the reddest provinces — pre-cap selection, not just post-cap repricing.

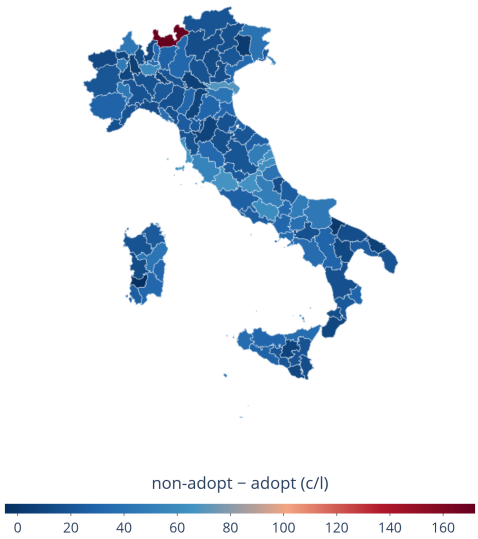


Figure 13: Petrol: pre-cap mean price of non-adopters minus adopters (c/l).

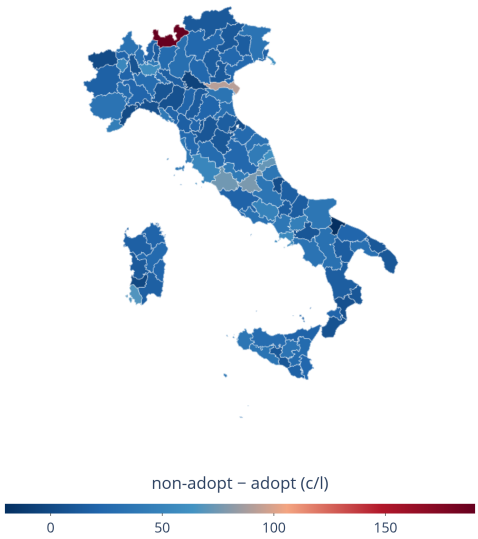


Figure 14: Diesel: pre-cap mean price of non-adopters minus adopters (c/l).

Post-cap price gap between non-adopters and adopters (28 September – 12 October). Non-adopters price above adopters in every province, by 58–258 c/l (petrol) and 60–282 c/l (diesel); the gap is reddest (widest) where adoption was high and the provincial mean is anchored at the cap.

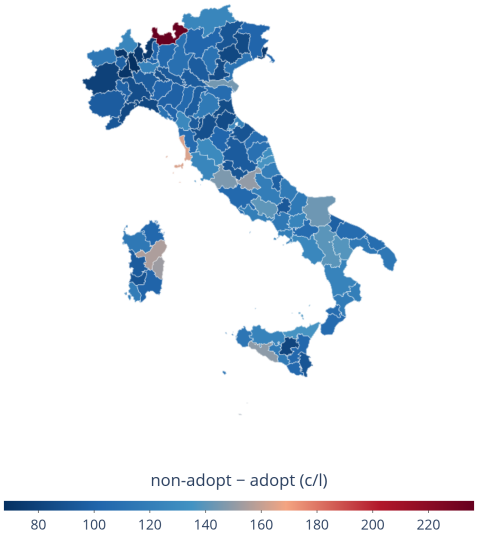


Figure 15: Petrol: mean post-cap price of non-adopters minus adopters (c/l).

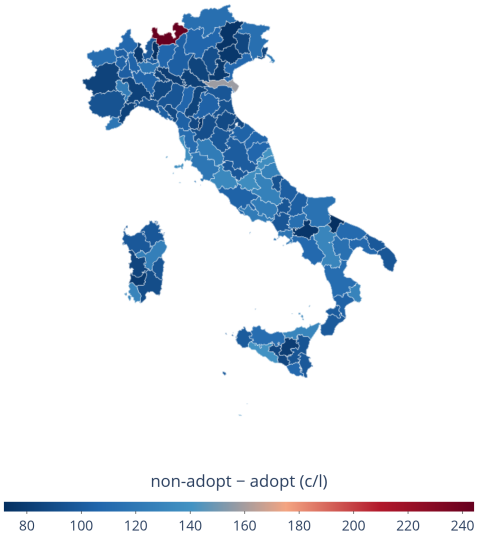


Figure 16: Diesel: mean post-cap price of non-adopters minus adopters (c/l).

Share of cap adopters 7 days after the cap. Adoption spans 32–74% of units across provinces in both fuels, with the reddest provinces concentrated in the central-northern corridors identified in fig. 3.

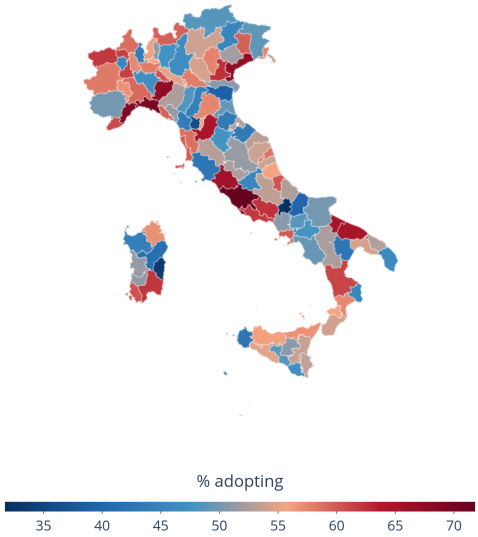


Figure 17: Petrol: share of units at or below the cap, 28 September – 5 October 2026 (%).

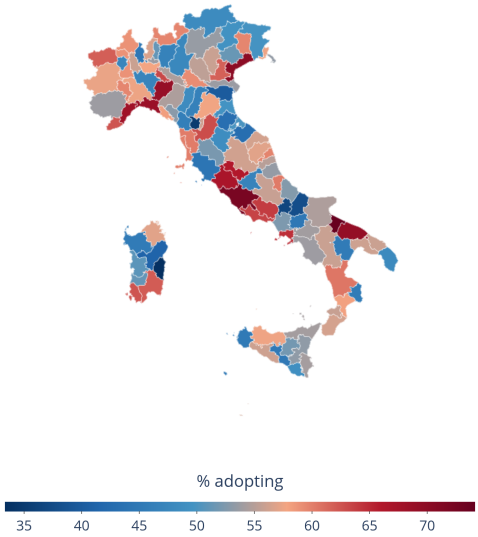


Figure 18: Diesel: share of units at or below the cap, 28 September – 5 October 2026 (%).

Dispersion and the non-adopter gap across provinces (main-text section 8). The four maps behind the distributions of fig. 6: within-province price SD before and after the cap, and the non-adopter gap before and after.

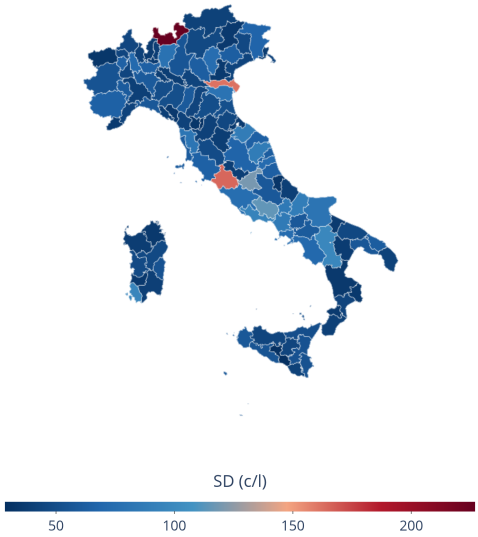


Figure 19: Within-province standard deviation of effective prices (c/l), pre-cap (21–27 September); mean across the two fuels. Red = high.

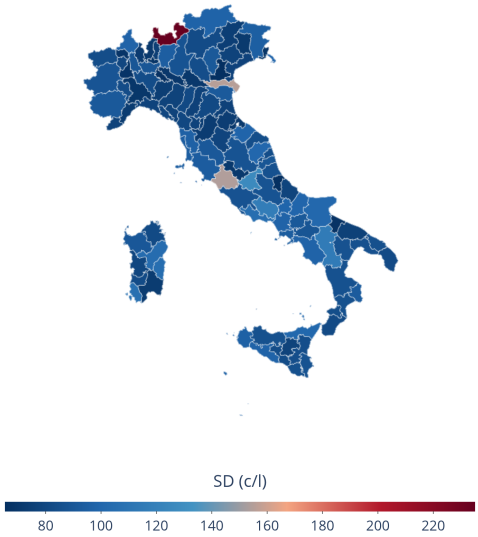


Figure 20: Within-province standard deviation of effective prices (c/l), post-cap (28 September – 12 October); mean across the two fuels. Red = high.

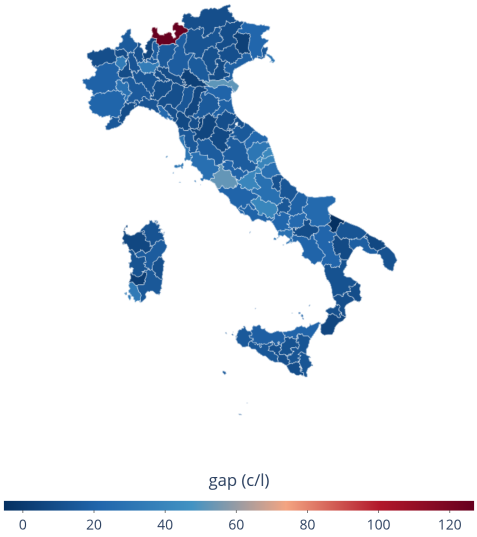


Figure 21: Non-adopters' mean gap to the provincial daily mean (c/l), pre-cap. Red = large gap.

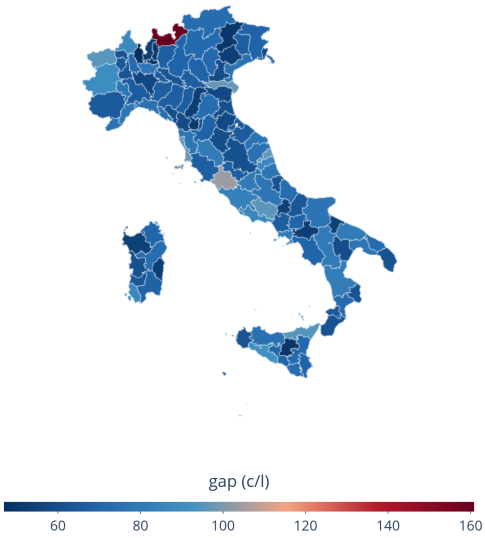


Figure 22: Non-adopters' mean gap to the provincial daily mean (c/l), post-cap (28 September – 12 October). Red = large gap.